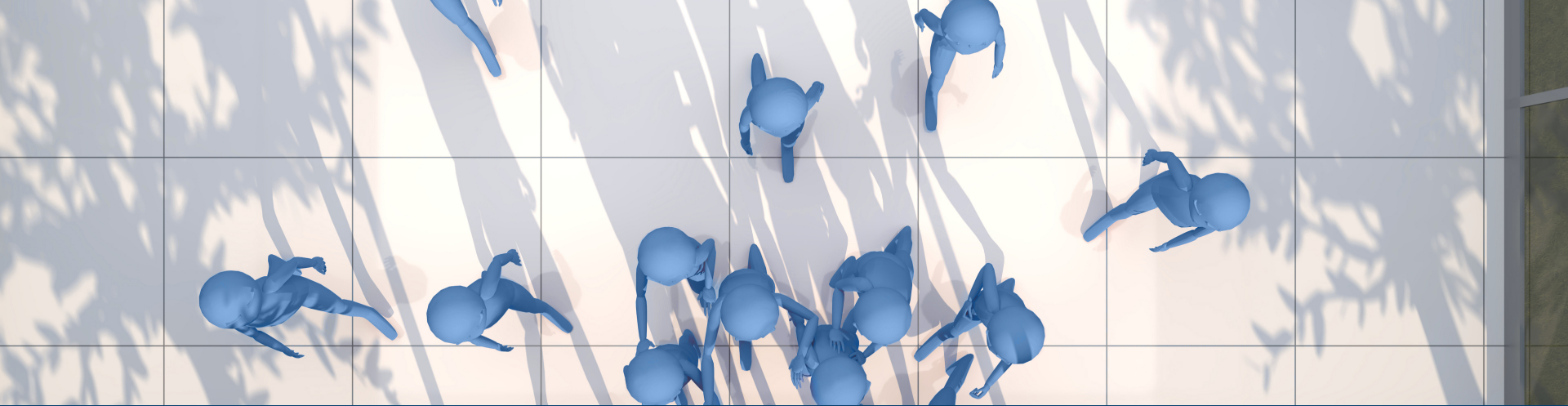




A MODULAR WORKFLOW FOR VISIBILITY-AWARE EVACUATION MODELLING

pyFDS-Evac: Integrating smoke effects into movement, route choice, and wayfinding

25 September 2026 | Mohcine Chraibi | Institute for Advanced Simulation, IAS-7 Civil Safety Research

Seminar on Visibility in Fire Safety Science and Engineering, Wuppertal

HUMAN BEHAVIOR IN FIRE

Definition?

ISO 13943:2023 defines human behaviour in fire as "actions performed in the event of a fire as a result of a behavioural or decision-making process".

1. Speed

How fast can I walk?

2. Routing

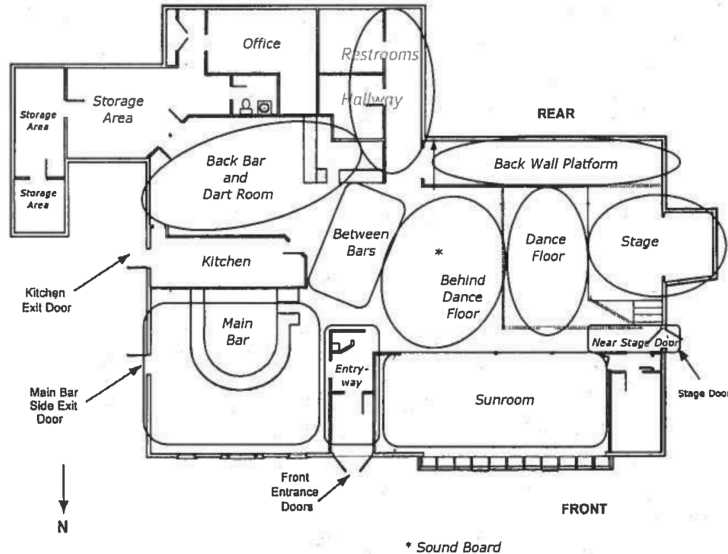
Which exit do I head for?

3. Wayfinding

Which exits do I even know about?

WHY BUILD THIS AT ALL

Gaps in Human Behavior in Fire



The Station Nightclub, West Warwick, 20 February 2003. Floor plan from NIST NCSTAR 2.

Four exits; most of the crowd went for the front entrance doors they had come in by.



PUBLICATIONS

Report of the Technical Investigation of The Station Nightclub Fire (NIST NCSTAR 2), Volume 1

Published: June 30, 2005

Author(s)

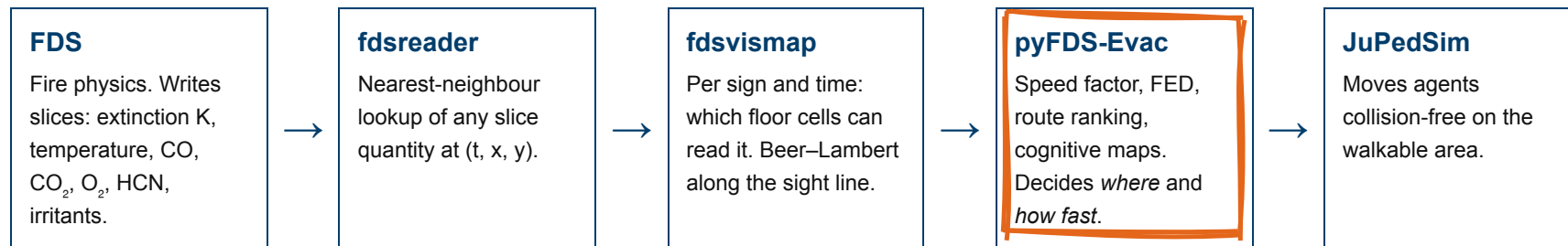
William L. Grosshandler, Nelson P. Bryner, Daniel M. Madrzykowski, K Kuntz

"One could argue that many of the people waiting to go out the front door would have chosen the barroom door ... The counter argument is that the barroom exit sign could have been obscured by smoke, making that door a reasonable alternative only for those familiar with the nightclub."

NIST NCSTAR 2 Vol. I, §6.6, The Station Nightclub fire

WHY NOW?

The necessary ingredients were not available at the time



We neither maintained nor rewrote FDS+Evac.

- Fortran inside the FDS source tree, one maintainer, FDS's release cycle
- When it died, none of the boxes above existed as a Python library
- A replacement meant writing a slice reader, a visibility model and a movement model ourselves

Python, and five packages instead of one.

- A model survives only if the community picks it up, works with it and extends it
- Everyone can read and change Python
- Each box is a small package with its own maintainers, tests and release cycle: responsibility is distributed, no single group carries the chain
- What remains for us is the glue, and the glue is small

FDS+EVAC AND PY-FDS-EVAC

Same empirical basis for smoke and dose. Different answers on route choice and knowledge.

	FDS+Evac	pyFDS-Evac
Smoke slows walking	Frantzich–Nilsson, floor $0.1 v_0$	Same law, same constants. Fridolf as an alternative
Smoke incapacitates	ISO 13571 FED from CO, CO ₂ , O ₂ . Stop at FED = 1	Twelve ISO 13571 species, convective heat as a second dose, per-agent threshold
Irritants	Not modelled	Purser's FIC slows walking instantly, recovers on leaving the plume
Smoke hides the way	Agents know every door. Familiarity is a fixed preference	Sign legibility through smoke (fdsvismap) decides which exits enter an agent's map. Staff know the building, visitors discover it

1 · SPEED

Visibility loss slows people down. The standard part, with one caveat about floors.

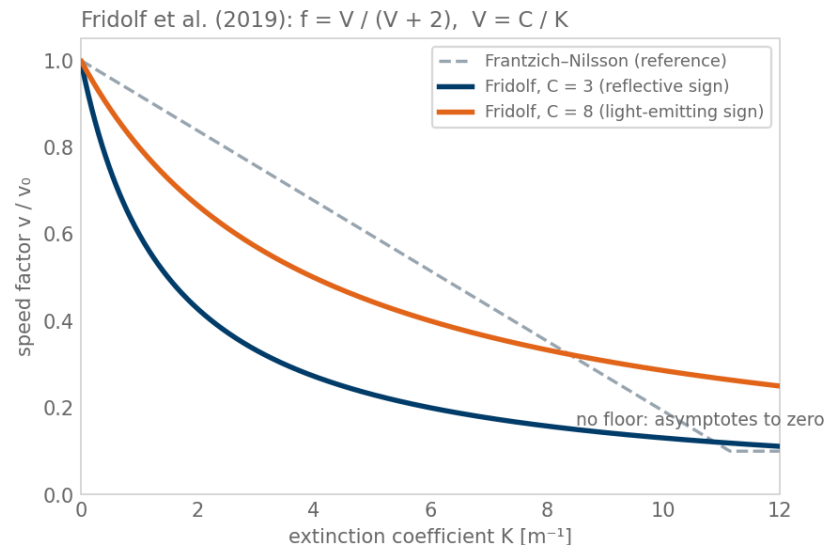
EXTINCTION TO WALKING SPEED

Two laws from the same kind of experiment, selectable in the code

Fridolf et al. (2019), via Jin's relation

$$V = \frac{C}{K}, \quad f(V) = \frac{V}{V+2}$$

- Calibrated on individual walking speeds in smoke-filled road and rail tunnels
- V is Jin's sighting distance; $C = 3$ for reflective signs, $C = 8$ for light-emitting signs
- Optional: implemented as `speed_law = "fridolf"`.



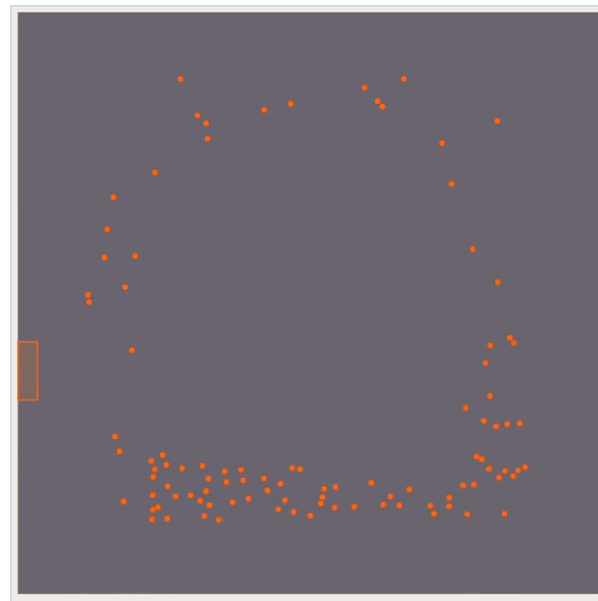
TENABILITY: A BRAKE AND A GATE

Speed varies smoothly with smoke and irritants. Dose only flips the switch.

$$v = v_0 \cdot f(K) \cdot g(\text{FIC}) \cdot 1[\text{FED} < 1]$$

- **Seeing**, $f(K)$: smoke density slows walking. A brake.
- **Breathing**, $g(\text{FIC})$: irritant gases slow walking, now. A brake that releases when the air clears.
- **Consciousness**, FED: toxic dose accumulates and, at the threshold, stops the agent. Heat dose is a second, independent gate.

Three independent systems, therefore multiplicative. Laws and constants as in FDS+Evac: Frantzich–Nilsson for f , Purser for FIC and FED. Thresholds per agent.



The brake acting: one room, exit at the left wall, smoke filling from the far side. Colour is cumulative FED, orange safe, red alert.

2 · ROUTE CHOICE

Smoke along a route, not at a point. Where the model departs from FDS+Evac.

ROUTES LIVE ON A STAGE GRAPH

Smoke is integrated along the corridor the agent will actually walk

- Directed graph $\mathcal{G} = (V, E)$: spawn areas, checkpoints, exits
- Each edge carries a **polyline** computed by JuPedSim's routing engine along the walkable area; weight L_{uv} is its arc length
- Path-mean extinction along the polyline, sample spacing $\Delta s \leq 2$ m:

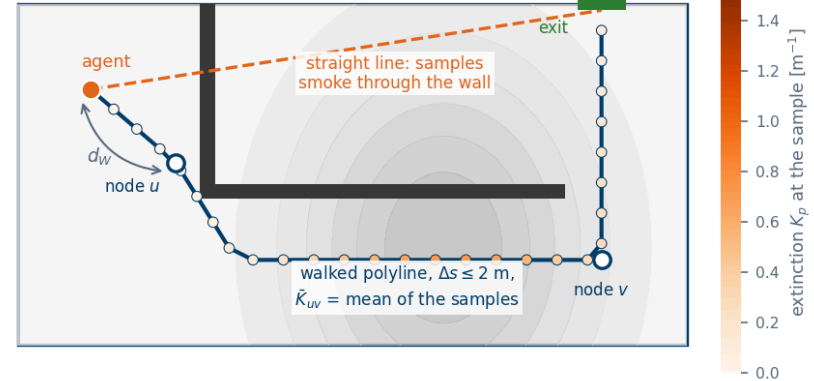
$$\bar{K}_{uv} = \frac{1}{|P|} \sum_{p \in P} K_p$$

the discrete Beer–Lambert mean of Börger et al. (2024), Eqs. 8–9, applied to a walked path instead of a sight line.

- Cost starts at the agent's **actual position** (d_w to the first node) and the route is **re-decided every T seconds**

FDS+Evac, JPSfire and B-RISK price smoke along a **straight line** from agent to door. In a non-convex plan that samples smoke through walls.

Plan view: cost is integrated along the walk, not the sight line



THE EXPOSURE GATE

One quantity refuses a route, then orders the survivors

One number per route: the smoke the agent would walk through

$$\tau_k = \bar{K}_k L_k \quad (\text{optical depth along the walk})$$

1. Refuse. $\tau_k > 6$: the route is out.

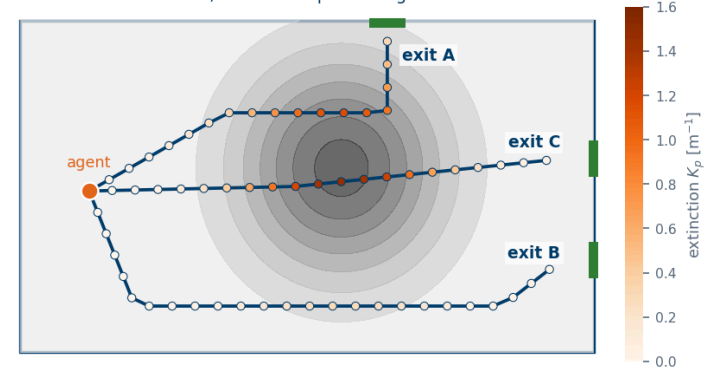
2. Rank the rest by τ_k . Cleanest route first. A long clean route beats a short smoky one.

3. Break ties by travel time R_k : the walking time along the route at the smoke-reduced speed $v_0 f(\bar{K})$. In clear air every $\tau_k = 0$ and R_k alone decides (shortest path).

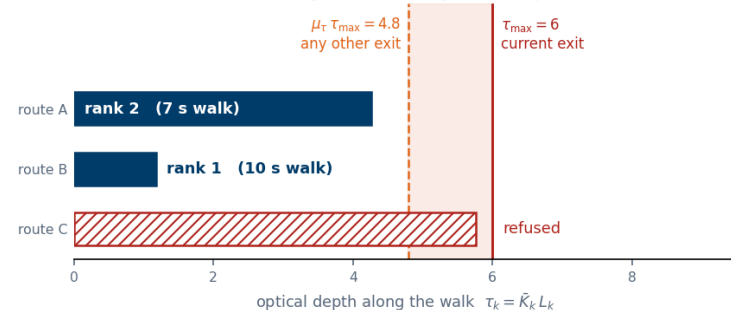
Projected FED at arrival > 1 is a separate veto: it removes an exit.

τ measures smoke **walked through**, not sighting distance. The budget 6 is FDS+Evac's door test (Jin, $c = 3$), inherited and uncalibrated as an exposure budget.

Three candidate routes, smoke sampled along each walk



Gate: refuse above the budget, then sort by τ , then by time



3 · WAYFINDING

What the agent knows. Sign legibility through smoke and per-agent cognitive maps.

A SIGN IS MORE THAN A POSITION

Legibility through smoke via fdsvismap

Exit and checkpoint signs are sign descriptors

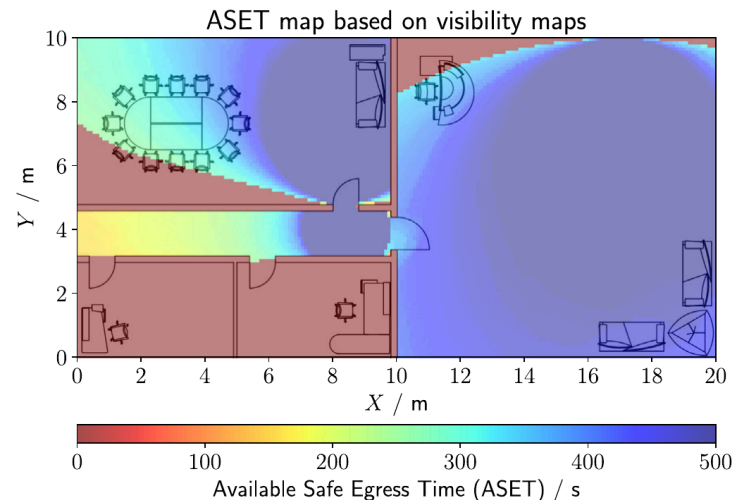
$$\{x, y, \alpha, c\}$$

- (x, y) position of the sign
- α viewing direction, compass bearing. A sign seen from behind is not seen
- c Beer–Lambert contrast parameter (Jin's c , in its proper role)

At runtime the model only asks

$$\text{visible}(t, \mathbf{x}_i, n) = \text{sign_is_visible}(t, x_i, y_i, \text{sign}(n))$$

evaluated from the agent's **actual position**, not the source node centroid. A table lookup, no ray marching in the loop.



Waypoint-based ASET map of an office (Börger, Belt, Arnold 2024, Fig. 11): time at which each cell loses sight of every exit sign. Rooms without a legible sign are red from the start, however clear the air.

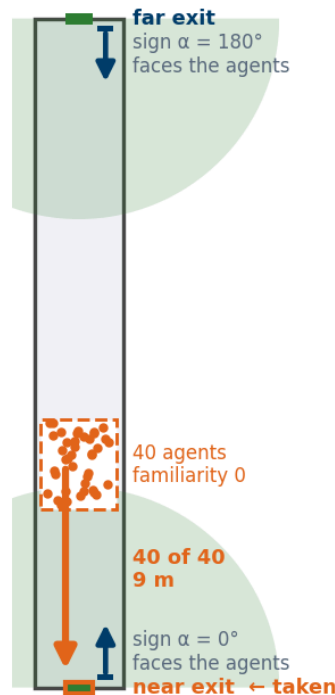
ONE NUMBER FLIPS THE EXIT

Turn the near exit's sign around and 40 agents walk twice as far

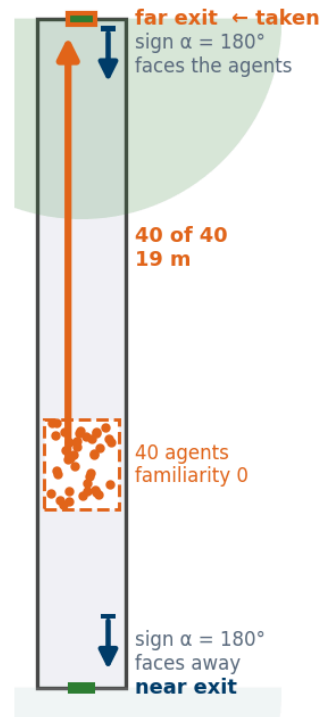
- The two configurations are identical except one number: the **bearing α** of the near exit's sign
- $\alpha = 0$, sign faces the agents: both exits enter the map, all 40 take the near one
- $\alpha = 180$, sign faces away: the near exit **never enters the map**, all 40 walk 19 m to the far one

The near exit is **absent, not rejected**. A rejected route can come back through the fallback; an unknown one cannot.

Near sign faces the agents

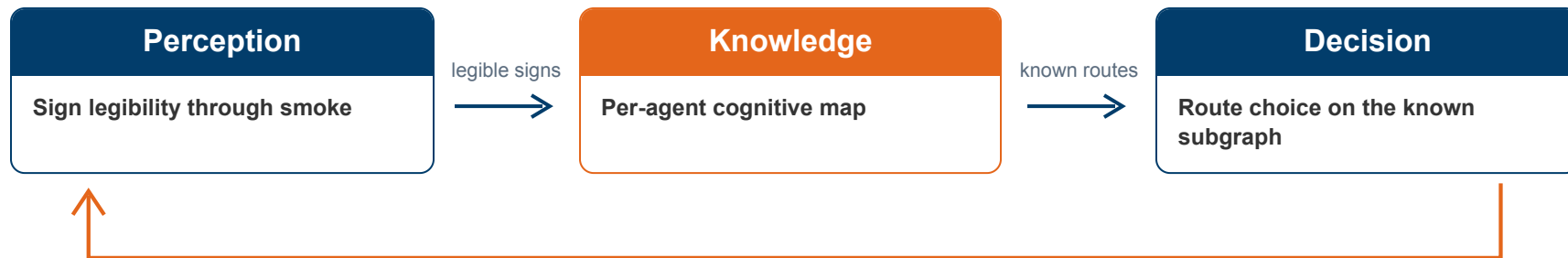


Near sign turned around



WHAT THE AGENT KNOWS

Perception feeds a map, the map feeds the decision, the walk feeds perception

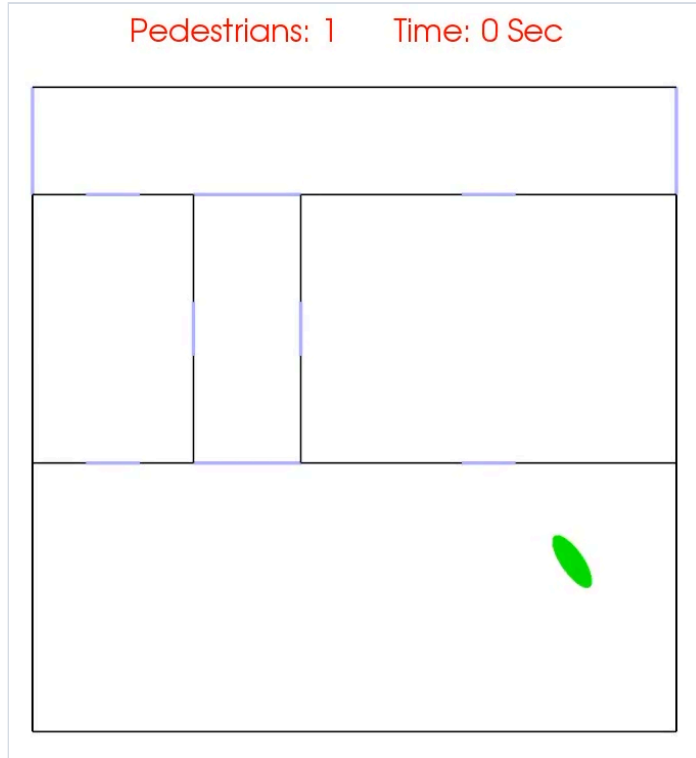


walking the chosen route brings new signs into view, and the map grows

The loop. Walking the chosen route brings new signs into view, so the map grows and the next decision is made on more knowledge. Sensing, a map, a planner: the classical robotics picture, with a visibility model as the sensor and a graph as the map.

AN EMPTY MAP

No sign is legible. Where does the agent go?



- One agent, **familiarity 0**, no legible sign anywhere in the building
- Its map holds a single node: where it stands. No exit, no door, no route
- Three plausible routes are drawn. Which one does the model take?

None of them. With no exit in the map there is nothing to route to. The fallback walks to the nearest known node and re-decides there, and with nothing new learned the answer is the same node. The agent circles.

PER-AGENT COGNITIVE MAPS

Staff know the building. Visitors discover it.

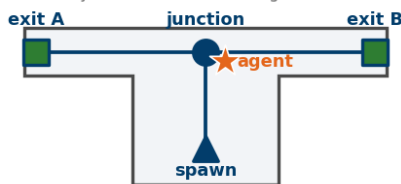
1 At spawn, $t = 0$

discovery: spawn + junction with sign in view



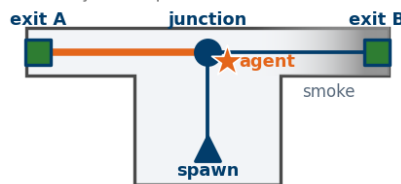
2 Arrives at the junction

discovery: both exits become legible



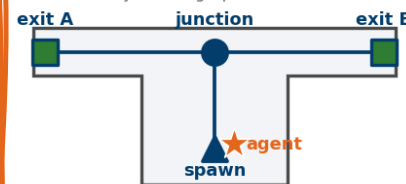
3 Smoke hides exit B

discovery: router picks exit A



4 At spawn, $t = 0$

full familiarity: whole graph known

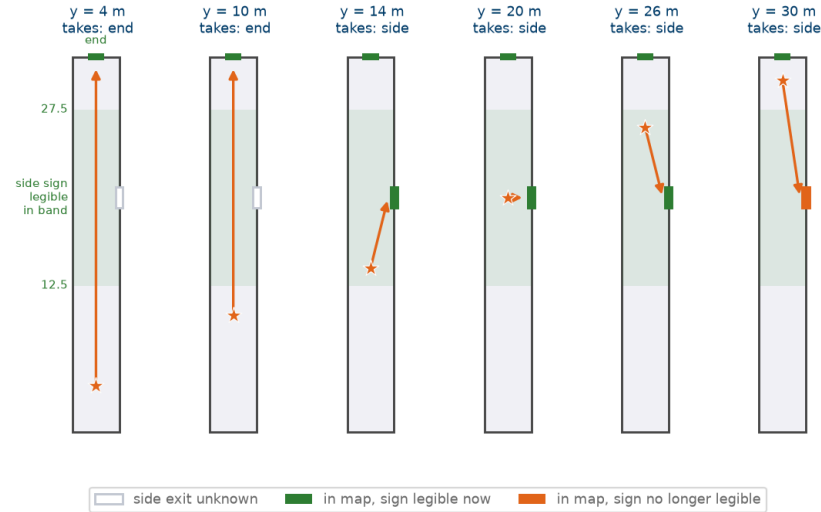


● known node ○ unknown node — known edge - - unknown edge — chosen route ★ agent

Full-familiarity agent, same spot, $t = 0$. The whole graph from the start: staff, or anyone who has been here before. Between the two tiers, a group may declare an entrance it always knows. After Hänsel (2014).

THE MAP REMEMBERS

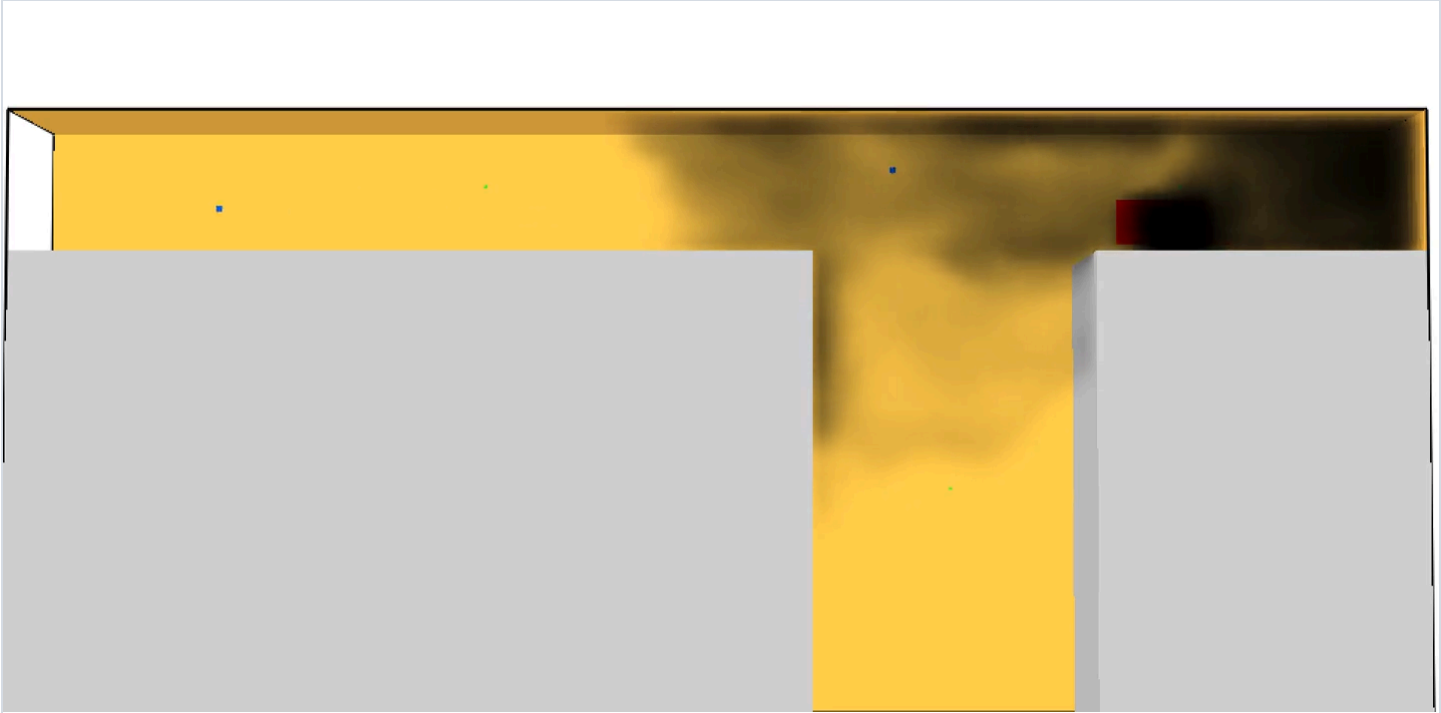
A sign seen once stays in the map after it can no longer be read



- **Acquisition:** the side exit enters the map at $y = 12.5$, and the agent routes to it from there
- **Persistence:** at $y = 30$ the sign is unreadable and the agent still routes to it. Memory outlives perception

THE T-CORRIDOR DEMO, RUN

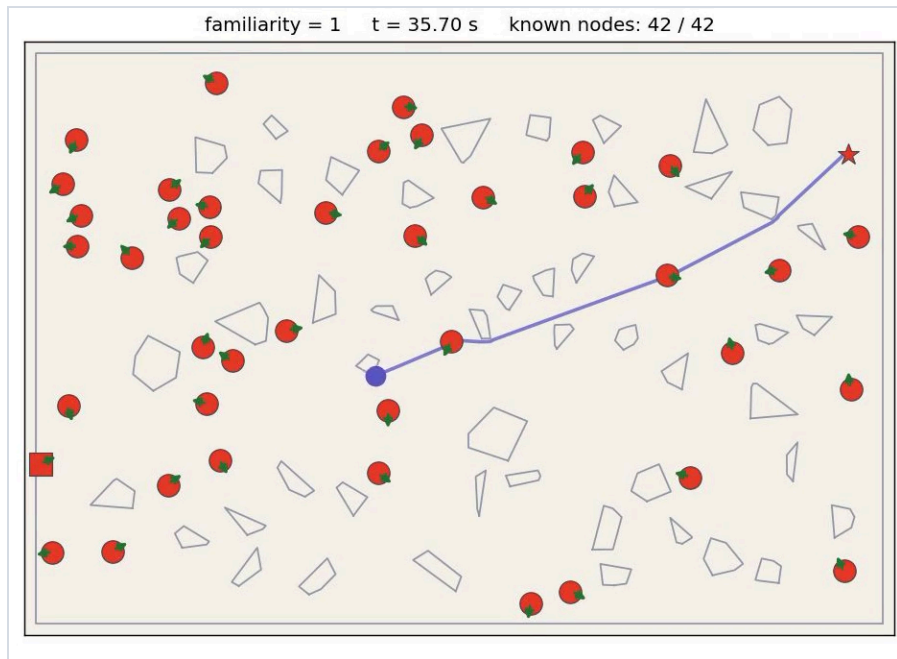
Signs fill the map, smoke prices the routes, the agents turn left



Coupled FDS run on the T-corridor of the previous slide, fire near exit B. Agents learn the junction from its sign and both exits from the junction. The exposure gate sends them to exit A once smoke fills the right arm.

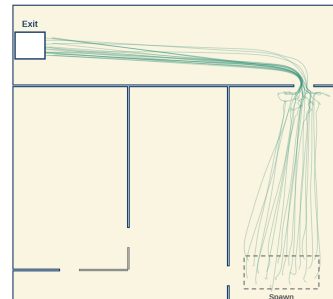
FULL VERSUS DISCOVERY, ON A GENERATED WORLD

Same seed, same 42 signs, same exit. Only the initial map differs.

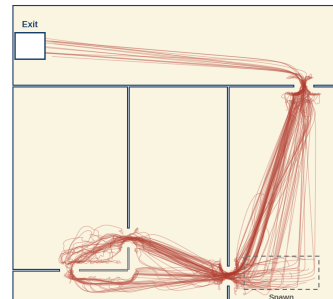


One agent, played at 2× speed. First pass: familiarity 1, all 42 nodes known at spawn, straight to the exit. Second pass: familiarity 0, the map grows only through signs that are legible from where the agent stands.

FULL



DISCOVERY



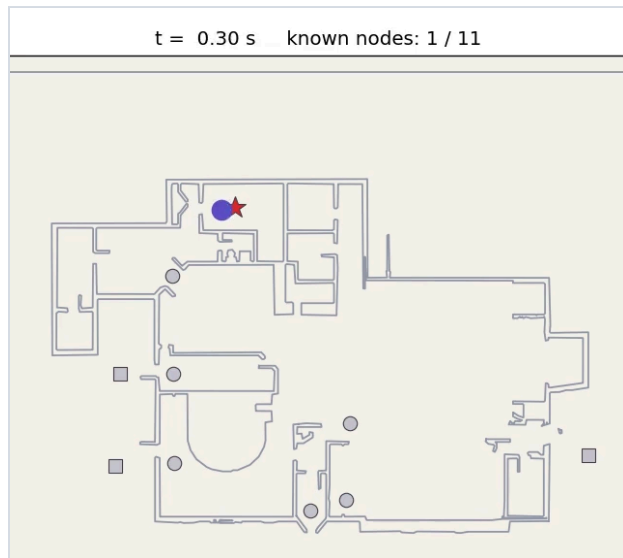
20 agents' walked paths

Twenty agents on a three-room deck. Full: 35.1 s to the last exit. Discovery: 75.1 s, with the detour into the dead-end room where nothing is legible.

THE STATION NIGHTCLUB, ONE VISITOR

Same floor plan, same spawn, same seed. Only what the agent can read differs.

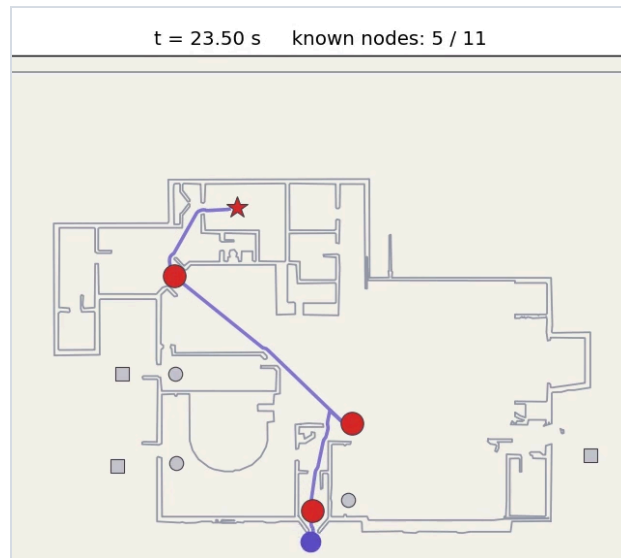
Discovery, no sign gating



Arriving at a node reveals all its graph neighbours. Eight of eleven nodes known after 6 s: the map grows with topology, not with sight.

Red: known exits (squares) and stages (circles). Grey: unknown. Star: spawn. Purple: agent and known edges. One agent, familiarity 0, NIST NCSTAR 2 floor plan, no fire. Frames are engine truth, recorded by `run_scenario`.

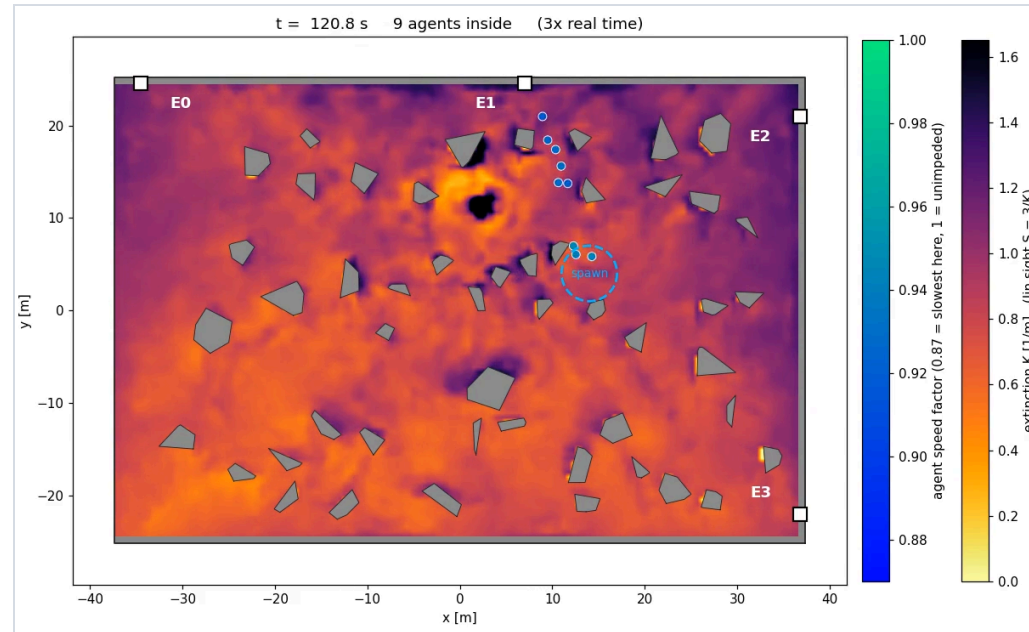
Discovery, sign-visibility gated (fdsvismap, clear air)



A neighbour enters the map only when its sign is legible from where the agent stands. Three of eleven nodes known after 14 s; the agent has to walk to learn.

SMOKE, SPEED AND EXIT CHOICE ON A GENERATED WORLD

Coupled FDS run: four exits, agents coloured by speed factor



Played at 3× real time. Background: FDS extinction coefficient K at 2 m, right colour bar also as J_{in} sighting distance $S = 3/K$. Dots: agents, coloured by the speed factor $f(K)$ they experience.

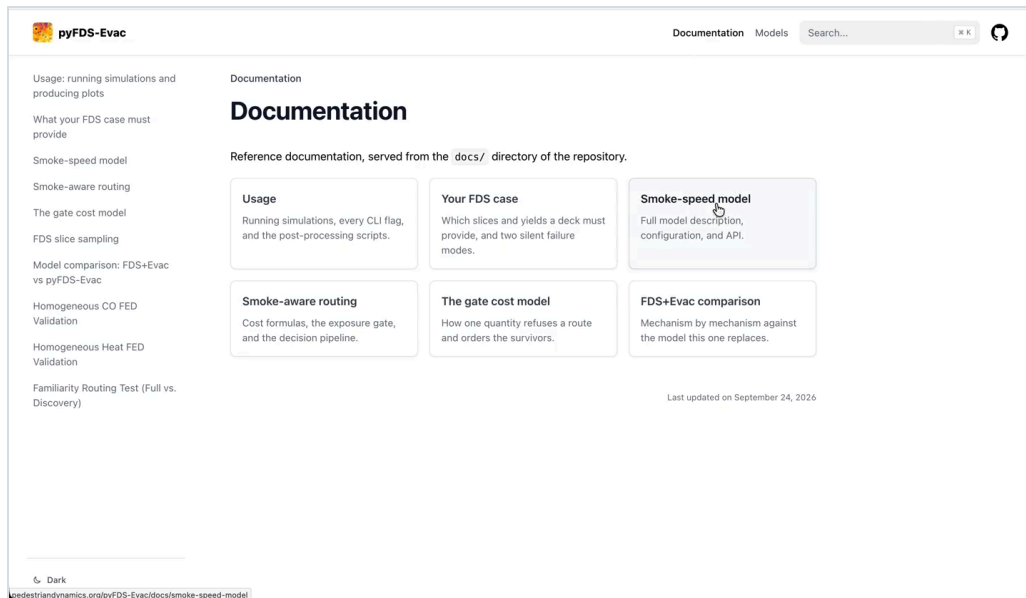
What the two colour bars say

- The smoke field is what the routing layer samples along every candidate polyline, at the time the agent would reach it
- The speed factor is what each agent actually walks with; on this fire it stays above 0.87, so smoke slows without stopping
- Exits E0 to E3 carry signs; the spawn circle is where every agent starts, so route choice is the only difference between them

Material slide. Which exits are chosen, and when the far ones are refused, is the thing to narrate over this clip.

DOCUMENTATION

pedestriandynamics.org/pyFDS-Evac

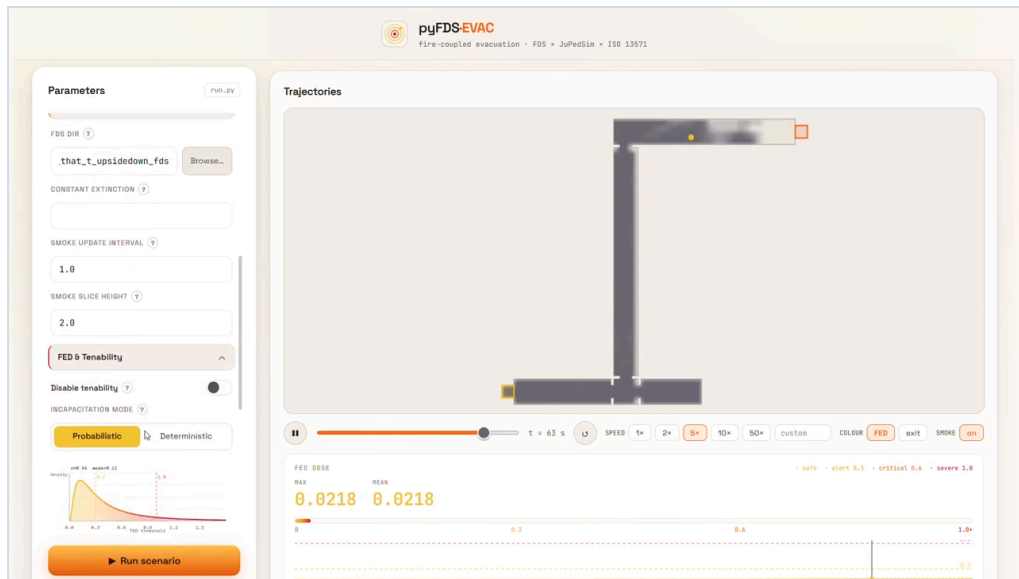


The documentation site, scrolled at double speed.

- **Usage:** every command-line flag, the post-processing scripts, the web GUI
- **Models:** the equations behind speed, dose, route choice and cognitive maps, with the configuration keys
- **Comparison:** FDS+Evac against pyFDS-Evac, mechanism by mechanism

BONUS: A BROWSER ON TOP OF THE LIBRARY

Not the focus. The library is the product; a web front end is cheap to build on it.



Screen recording of the web GUI on a T-shaped test deck. Agents coloured by cumulative FED, the FDS extinction slice drawn underneath as a smoke layer.

- The deliverable is a lightweight Python library:
`pip install` , import, call `run_scenario()`
from a script, a notebook or a test
- The GUI is a thin convenience layer on top: every form field is a flag of the command-line runner, and it calls the same `run_scenario()`
- Replay: agents coloured by FED or by assigned exit; the same output opens in Smokeview next to the fire

Because the model has exactly one entry point, a front end can be built on the library without touching it. This one is the work of **Gregory Shouldice**.

TAKE-HOME

Modular

`FDS`, `fdsreader`, `fdsvismap` and `JuPedSim` each do one thing.
`pyFDS-Evac` adds no fire model and no movement model, only the decisions in between.

Code and paper

- Repo: github.com/PedestrianDynamics/pyFDS-Evac
- Docs: pedestriandynamics.org/pyFDS-Evac
- Paper: *pyFDS-Evac: A Modular Successor to FDS+Evac*, in preparation

Visibility-aware

Smoke slows agents through $f(K)$, refuses and orders routes through a path-integrated optical depth, and decides which exits an agent learns through sign legibility from `fdsvismap`. Legibility shapes knowledge, not permission.

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